

Model Predictive Control

Chapter 9: Robust MPC for Linear Systems Part II

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Spring 2026

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Goals of Robust Constrained Control

Uncertain constrained linear system

$$x(k+1) = Ax(k) + Bu(k) + w(k) \quad x, u \in \mathcal{X}, \mathcal{U} \quad w \in \mathcal{W}$$

Design control law $u(k) = \kappa(x(k))$ such that the system:

1. Satisfies constraints : $\{x(k)\} \subset \mathcal{X}$, $\{u(k)\} \subset \mathcal{U}$ for all disturbance realizations
2. Is stable: Converges to a neighbourhood of the origin
3. Optimizes (expected/worst-case) “performance”
4. Maximizes the set $\{x(0) \mid \text{Conditions 1-3 are met}\}$

Challenge: Cannot predict where the state of the system will evolve

We can only compute a set of trajectories that the system *may* follow

Idea: Design a control law that will satisfy constraints and stabilize the system *for all possible disturbances*

Learning Objectives

- Understand benefit of closed-loop predictions in presence of uncertainties
- Understand and contrast concepts of constraint-tightening and tube MPC
- Design tube MPC problem
- Prove robust constraint satisfaction and robust stability of tube MPC

Outline

1. Closed-Loop Predictions
2. Error dynamics
3. Robust Constraint-tightening MPC
4. Robust Tube MPC

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MPC as a Game

Two players: Controller vs Disturbance

$$x(k+1) = g(x(k), u(k)) + w(k)$$

1. Controller chooses his move u
2. Disturbance decides on his move w **after seeing the controller's move**

MPC as a Game

Two players: Controller vs Disturbance

$$x(k+1) = g(x(k), u(k)) + w(k)$$

1. Controller chooses his move u
2. Disturbance decides on his move w **after seeing the controller's move**

What are we assuming when making open-loop robust predictions?

1. Controller chooses a **sequence** of N moves in the future $\{u_0, \dots, u_{N-1}\}$
2. Disturbance chooses N moves **knowing all N moves of the controller**

We are assuming that the controller will do the same thing in the future no matter what the disturbance does!

Can we do better?

Closed-Loop Predictions

What should the future prediction look like?

1. Controller decides his first move u_0
2. Disturbance chooses his first move w_0
3. Controller decides his second move $u_1(x_1)$ **as a function of the first disturbance** w_0 (**recall** $x_1 = Ax_0 + Bu_0 + w_0$)
4. Disturbance chooses his second move w_1 as a function of u_1
5. Controller decides his second move $u_2(x_2)$ **as a function of the first two disturbances** w_0, w_1
6. ...

Closed-Loop Predictions

We want to optimize over a **sequence of functions** $\{u_0, \mu_1(\cdot), \dots, \mu_{N-1}(\cdot)\}$, where $\mu_i(x_i) : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called a **control policy**, and maps the state at time i to an input at time i .

Notes:

- This is the same as making μ a function of the disturbances to time i , since the state is a function of the disturbances up to that point
- The first input u_0 is a function of the current state, which is known. Therefore it is not a function, but a single value.

The problem: We can't optimize over arbitrary functions!

Closed-Loop MPC

A solution: Assume some structure on the functions μ_i

Pre-stabilization $\mu_i(x_i) = Kx_i + v_i$

- Fixed K , such that $A + BK$ is stable
- Simple, often conservative

Linear feedback $\mu_i(x_i) = K_i x_i + v_i$

- Optimize over K_i and v_i
- Non-convex. Extremely difficult to solve...

Disturbance feedback $\mu_i(x_i) = \sum_{j=0}^{i-1} M_{ij} w_j + v_i$

- Optimize over M_{ij} and v_i
- Equivalent to linear feedback, but convex!
- Can be very effective, but computationally intense.

Tube MPC $\mu_i(x_i) = v_i + K(x_i - \bar{x}_i)$

- Fixed K , such that $A + BK$ is stable
- Optimize over $\bar{x}_i$ and v_i
- Simple, and can be effective

We will cover robust constraint-tightening and tube MPC in this lecture.

Outline

1. Closed-Loop Predictions
2. Error dynamics
3. Robust Constraint-tightening MPC
4. Robust Tube MPC

Main Idea

$$x(k+1) = Ax(k) + Bu(k) + w(k) \quad x, u \in \mathcal{X} \times \mathcal{U} \quad w \in \mathcal{W}$$

The idea: Separate the available control authority into two parts

1. A portion that steers the noise-free ‘nominal’ system to the origin
 $z(k+1) = Az(k) + Bv(k)$
2. A portion that compensates for deviations from this system, i.e. a ‘tracking’ controller, to keep the real trajectory close to the nominal

$$u_i = K(x_i - z_i) + v_i$$

for some linear controller K , which stabilizes the nominal system.

⇒ We fix the linear feedback controller K offline, and optimize over the nominal inputs $\{v_0, \dots, v_{N-1}\}$ and nominal trajectory $\{z_0, \dots, z_N\}$, which results in a convex problem.

Error Dynamics

- Define the error $e_i = x_i - z_i$ and feedback $u_i = v_i + K(x_i - z_i)$
This results in the error dynamics:

$$\begin{aligned}e_{i+1} &= x_{i+1} - z_{i+1} \\&= Ax_i + Bu_i + w_i - Az_i - Bv_i \\&= Ax_i + BK(x_i - z_i) + Bv_i + w_i - Az_i - Bv_i \\&= (A + BK)(x_i - z_i) + w_i \\&= (A + BK)e_i + w_i\end{aligned}$$

Note: Dynamics $A + BK$ are stable, and the set $\mathcal{W}$ is bounded, so there is some set $\mathcal{E}$ that e will stay inside for all time.

- Goal: Bound maximum error, or how far the 'real' trajectory is from the nominal

Uncertain State Evolution

Consider the system $e_{i+1} = A_K e_i + w_i$ and assume that $e_0 = 0$.

Where can the error evolve to? (i.e., how close can we stay to the origin?)

$$e_1 = w_0$$

$$e_2 = A_K e_1 + w_1 = A_K w_0 + w_1$$

$$\vdots$$

$$e_i = \sum_{j=0}^{i-1} A_K^j w_{i-1-j}$$

Assume that $w_i \in \mathcal{W}$ for all i . What is the set $\mathcal{F}_i$ that contains all possible states e_i ?

$$\mathcal{F}_i = \mathcal{W} \oplus A_K \mathcal{W} \dots \oplus A_K^{i-1} \mathcal{W} = \bigoplus_{j=0}^{i-1} A_K^j \mathcal{W}, \quad \mathcal{F}_0 := \{0\}$$

where $P \oplus Q := \{x + y \mid x \in P, y \in Q\}$

Minimum Robust Invariant Set

As sum goes to infinity, we arrive at the **minimum robust invariant set** mRPI

$$\mathcal{F}_\infty = \bigoplus_{j=0}^{\infty} A_K^j \mathcal{W}, \quad \mathcal{F}_0 := \{0\}$$

If there exists an n such that $\mathcal{F}_n = \mathcal{F}_{n+1}$, then $\mathcal{F}_n = \mathcal{F}_\infty$

Minimal Robust Invariant Set

Input: A_K

Output: $\mathcal{F}_\infty$

$\Omega_0 \leftarrow \{0\}$

loop

$\Omega_{i+1} \leftarrow \Omega_i \oplus A_K^i \mathcal{W}$

if $\Omega_{i+1} = \Omega_i$ **then**

return $\mathcal{F}_\infty = \Omega_i$

end if

end loop

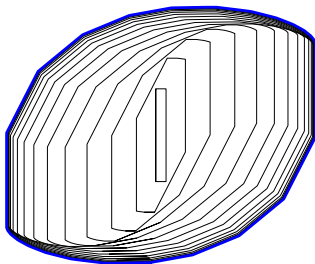
- A finite n does not always exist, but a 'large' n is a good approximation
- If n is not finite, there are other methods of computing small invariant sets, which will be slightly larger than $\mathcal{F}_\infty$

Example

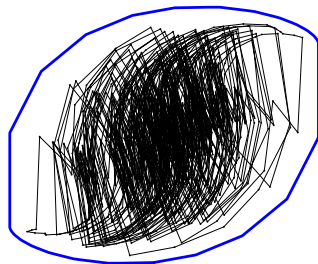
$$x(k+1) = \left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} K \right) x(k) + w(k)$$

$$\mathcal{W} := \{w \mid |w_1| \leq 0.01, |w_2| \leq 0.1\}$$

where K is the LQR controller for $Q = I$, $R = 10$.



Sets $A^i \mathcal{W}$ converging to minimal robust invariant set $\mathcal{F}_\infty$ in the limit



The state trajectory will stay in the set $\mathcal{F}_\infty$ for all time

From Error Bounds to Robust MPC - Two Variants

Consider error dynamics $e_{i+1} = (A + BK)e_i + w_i = A_K e_i + w_i \quad w_i \in \mathcal{W}$

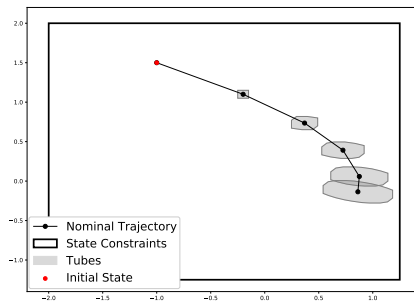
Disturbance reachable sets (DRS):

$$e_i \in \mathcal{F}_i = \bigoplus_{j=0}^{i-1} A_K^j \mathcal{W}, \text{ if } e_0 = 0$$

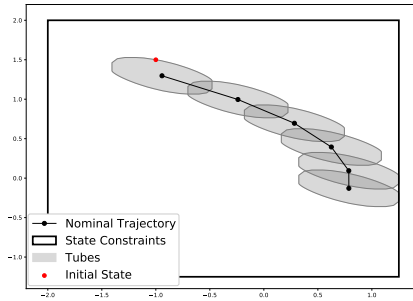
Robust positively invariant (RPI) set:

$$e_i \in \mathcal{E} \Rightarrow e_{i+1} \in \mathcal{E}$$

for any $e_0 \in \mathcal{E}$, we have $e_i \in \mathcal{E}$



→ Constraint-tightening MPC



→ Tube MPC

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Noisy System Trajectory

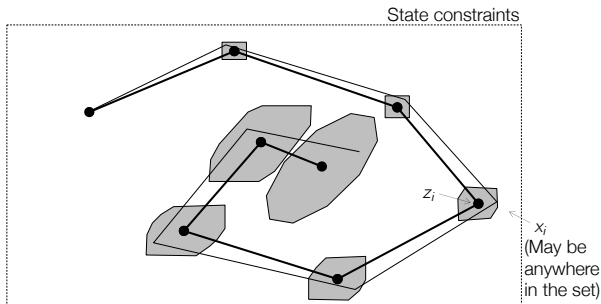
Given the nominal trajectory z_i , what can the noisy system trajectory do?

$$x_i = z_i + e_i$$

Don't know what error will be at time i , but it will be in the set $\mathcal{F}_i$, where $\mathcal{F}_i := \mathcal{W} \oplus A_K \mathcal{W} \oplus \dots \oplus A_K^{i-1} \mathcal{W} = \bigoplus_{j=0}^{i-1} A^j \mathcal{W}$, $\mathcal{F}_0 := \{0\}$.

Therefore, x_i can only be up to $\mathcal{F}_i$ far from z_i

$$x_i \in z_i \oplus \mathcal{F}_i = \{z_i + e_i \mid e_i \in \mathcal{F}_i\}$$



Constraint Tightening - Disturbance Reachable Sets

Goal: $x_i, u_i \in \mathcal{X} \times \mathcal{U}$ for all $\{w_0, \dots, w_{i-1}\} \in \mathcal{W}^i$

We want to work with the nominal system $z(k+1) = Az(k) + Bv(k)$ but ensure that the noisy system $x(k+1) = Ax(k) + Bu(k) + w(k)$ satisfies the constraints.

Necessary and sufficient condition:

$$z_i \oplus \mathcal{F}_i \subseteq \mathcal{X} \quad \Leftrightarrow \quad z_i \in \mathcal{X} \ominus \mathcal{F}_i$$

The sets $\mathcal{F}_i$ are known offline - we can compute the constraints $\mathcal{X} \ominus \mathcal{F}_i$ offline!

A similar condition holds for the inputs:

$$u_i \in v_i \oplus K\mathcal{F}_i \subseteq \mathcal{U} \quad \Leftrightarrow \quad v_i \in \mathcal{U} \ominus K\mathcal{F}_i$$

Robust Constraint-Tightening MPC Problem

$$\min_{Z, V} \sum_{i=0}^{N-1} l(z_i, v_i) + l_f(z_N)$$

$$\text{subj. to } z_{i+1} = Az_i + Bv_i$$

$$z_i \in \mathcal{X} \ominus \mathcal{F}_i$$

$$v_i \in \mathcal{U} \ominus K\mathcal{F}_i$$

$$z_N \in \mathcal{X}_f^{\text{ct}} \ominus \mathcal{F}_N$$

$$z_0 = x(k)$$

- Initial condition $z_0 = x(k)$
- Applied control: $u(k) = v_0^* + K(x(k) - z_0^*) = v_0^*$
- Disturbance reachable sets $\mathcal{F}_i := \mathcal{W} \oplus A_K \mathcal{W} \oplus \dots \oplus A_K^{i-1} \mathcal{W}$, $\mathcal{F}_0 := \{0\}$
- Terminal set $\mathcal{X}_f^{\text{ct}}$ robust invariant

Robust Constraint-Tightening MPC - Properties

Robust Invariance

Suppose the terminal ingredients $(l_f, \mathcal{X}_f^{\text{ct}})$ are designed such that:
 $\mathcal{X}_f^{\text{ct}} \subseteq \mathcal{X}$ and for all $z \in \mathcal{X}_f^{\text{ct}}$

- $Kz \in \mathcal{U}$
- $(A + BK)z + w \in \mathcal{X}_f^{\text{ct}} \forall w \in \mathcal{W}$

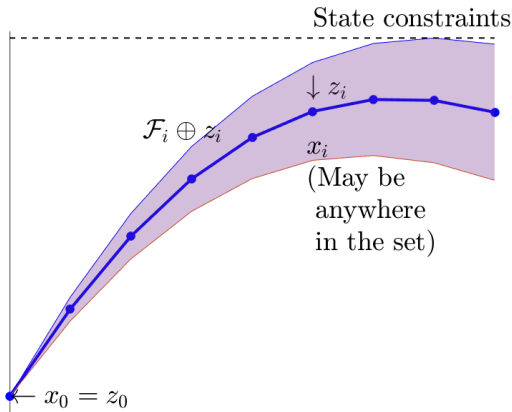
Let $\mathcal{X}_N$ be the feasible set and $V^*(x(k))$ be the optimizer of the robust constraint-tightening MPC problem for $x(k) \in \mathcal{X}_N$. Then $Ax(k) + Bv_0^*(x(k)) + w(k) \in \mathcal{X}_N$ for all $w \in \mathcal{W}$, i.e. the problem is recursively feasible.

This follows because the trajectory we computed at the current time is feasible for *any* disturbance, and therefore it's feasible for the one that we actually observe.

Note: Stability analysis needs a different concept (input-to-state stability)

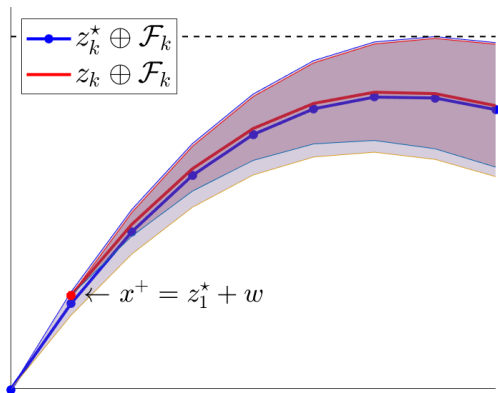
→ Theory in Advanced MPC

Constraint-tightening MPC - intuition



- Disturbance reachable set (DRS) $\mathcal{F}_i$ around nominal trajectory

Constraint-tightening MPC - intuition



- Disturbance reachable set (DRS) $\mathcal{F}_i$ around nominal trajectory
- Nominal candidate perturbed due to disturbance
- DRS around candidate prediction contained in previous DRS.

Constraint-tightening MPC - Proof Sketch

- Consider the candidate solution based on the error feedback and terminal controller:

$$v_i = v_{i+1}^* + K(z_i - z_{i+1}^*), \quad i = 0, \dots, N-1, \quad v_N^* = Kz_N^*, \quad z_{N+1}^* = A_K z_N^*.$$

- Recursive application yields:

$$z_i = z_{i+1}^* + A_K^i w(k), \quad i = 0, \dots, N.$$

$$v_i = v_{i+1}^* + K A_K^i w(k), \quad i = 0, \dots, N-1.$$

Induction proof: $z_{i+1} = Az_i + Bv_i =$
 $A(z_{i+1}^* + A_K^i w(k)) + B(v_{i+1}^* + K(z_i - z_{i+1}^*)) = z_{i+2}^* + A_K^{i+1} w(k).$

- Nested prediction:

$$z_i \oplus \mathcal{F}_i = z_{i+1}^* + A_K^i w(k) \oplus \mathcal{F}_i \subseteq z_{i+1}^* \oplus A_K^i \mathcal{W} \oplus \mathcal{F}_i = z_{i+1}^* \oplus \mathcal{F}_{i+1}.$$

Constraint-tightening MPC - Proof Sketch (cont.)

- Nested prediction: $z_i \oplus \mathcal{F}_i \subseteq z_{i+1}^* \oplus \mathcal{F}_{i+1}$.

- Terminal constraint:

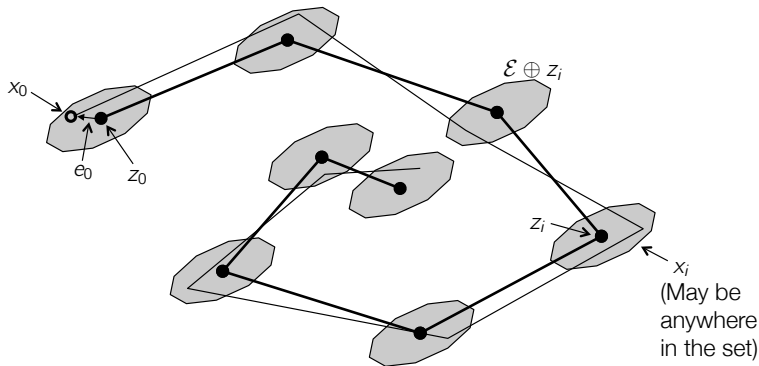
$$z_N \oplus \mathcal{F}_N \subseteq z_{N+1}^* \oplus \mathcal{F}_{N+1} = A_K(z_N^* \oplus \mathcal{F}_N) \oplus \mathcal{W} \subseteq A_K \mathcal{X}_f \oplus \mathcal{W} \stackrel{\text{RPI}}{\subseteq} \mathcal{X}_f.$$

- Feasible candidate ensures recursive feasibility.
- Constraint satisfaction with $x = z_0^* \in \mathcal{X}$, $u = v_0^* \in \mathcal{U}$.

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Tube MPC : The Idea

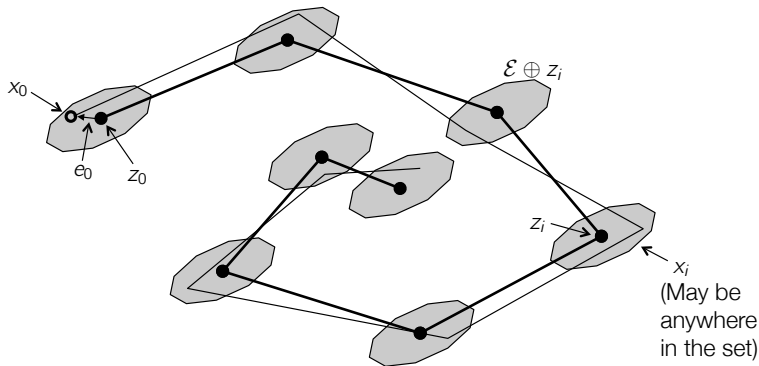


We ignore the noise and plan the **nominal trajectory**.

Bound the maximum error at any time with RPI set $\mathcal{E}$: $e_i \in \mathcal{E} \Rightarrow e_{i+1} \in \mathcal{E}$.

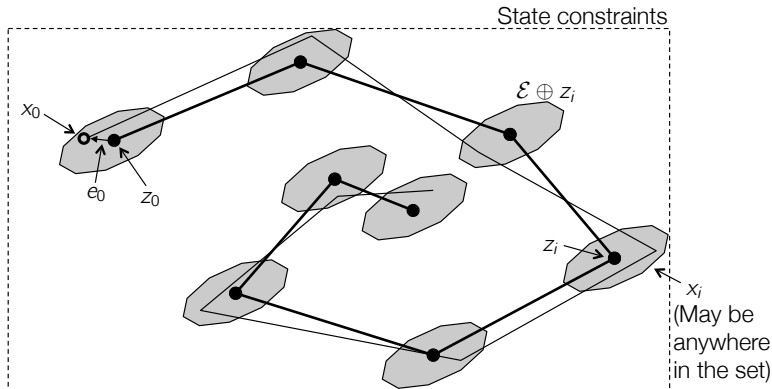
Ideally $\mathcal{E}$ is selected as the minimum RPI set $\mathcal{F}_\infty = \bigoplus_{j=0}^{\infty} A^j \mathcal{W}$.

Tube MPC : The Idea



We know that the real trajectory stays 'nearby' the nominal one: $x_i \in z_i \oplus \mathcal{E}$
because we plan to apply the controller $u_i = K(x_i - z_i) + v_i$ in the future
(we won't actually do this, but it's a valid sub-optimal plan)

Tube MPC : The Idea



We must ensure that all possible state trajectories satisfy the constraints
This is now equivalent to ensuring that $z_i \oplus \mathcal{E} \subset \mathcal{X}$
(address input constraints later)

Tube MPC

What do we need to make this work?

- Compute the set $\mathcal{E}$ that the error will remain inside
- Modify constraints on nominal trajectory $\{z_i\}$
- Formulate as convex optimization problem

... and then prove that

- Constraints are robustly satisfied
- The closed-loop system is robustly stable

Tube MPC

What do we need to make this work?

- **Compute the set $\mathcal{E}$ that the error will remain inside**
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Recall: Robust Invariant Set

Robust constraint satisfaction, for an **autonomous** system

$x(k+1) = g(x(k), w(k))$, or **closed-loop** system

$x(k+1) = g(x(k), \kappa(x(k)), w(k))$ for a **given** controller κ .

Robust Positive Invariant set

A set $\mathcal{O}^{\mathcal{W}}$ is said to be a robust positive invariant set for the autonomous system $x(k+1) = g(x(k), w(k))$ if

$$x \in \mathcal{O}^{\mathcal{W}} \Rightarrow g(x, w) \in \mathcal{O}^{\mathcal{W}}, \text{ for all } w \in \mathcal{W}$$

Previously we wanted the **maximum robust invariant set**, or the largest set in which our terminal control law works.

We now want the **minimum robust invariant set**, or the smallest set that the state will remain inside despite the noise.

Tube MPC

What do we need to make this work?

- Compute the set $\mathcal{E}$ that the error will remain inside
- **Modify constraints on nominal trajectory** $\{z_i\}$
- Formulate as convex optimization problem

... and then prove that

- Constraints are robustly satisfied
- The closed-loop system is robustly stable

Noisy System Trajectory - Constant Tube

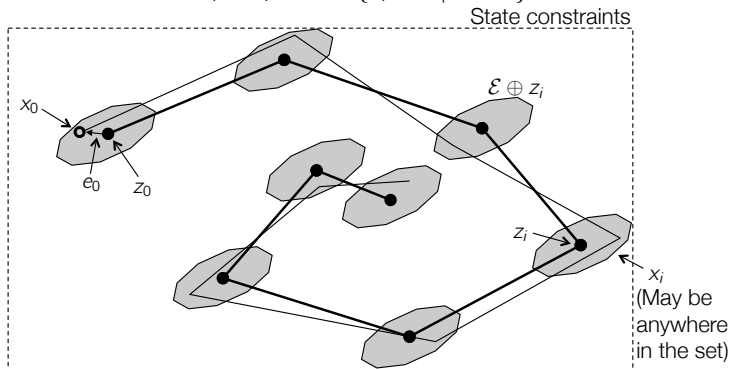
Given the nominal trajectory z_i , what can the noisy system trajectory do?

$$x_i = z_i + e_i$$

Don't know what error will be at time i , but it will be in the set $\mathcal{E}$

Therefore, x_i can only be up to $\mathcal{E}$ far from z_i

$$x_i \in z_i \oplus \mathcal{E} = \{z_i + e \mid e \in \mathcal{E}\}$$



Constraint Tightening

Goal: $x_i, u_i \in \mathcal{X} \times \mathcal{U}$ for all $\{w_0, \dots, w_{i-1}\} \in \mathcal{W}^i$

We want to work with the nominal system $z(k+1) = Az(k) + Bv(k)$ but ensure that the noisy system $x(k+1) = Ax(k) + Bu(k) + w(k)$ satisfies the constraints.

Necessary and sufficient condition:

$$z_i \oplus \mathcal{E} \subseteq \mathcal{X} \quad \Leftrightarrow \quad z_i \in \mathcal{X} \ominus \mathcal{E}$$

The set $\mathcal{E}$ is known offline - we can compute the constraints $\mathcal{X} \ominus \mathcal{E}$ offline!

A similar condition holds for the inputs:

$$u_i \in K\mathcal{E} \oplus v_i \subset \mathcal{U} \quad \Leftrightarrow \quad v_i \in \mathcal{U} \ominus K\mathcal{E}$$

Tube MPC

What do we need to make this work?

- Compute the set $\mathcal{E}$ that the error will remain inside
- Modify constraints on nominal trajectory $\{z_i\}$
- **Formulate as convex optimization problem**

... and then prove that

- Constraints are robustly satisfied
- The closed-loop system is robustly stable

Tube MPC Problem Formulation

$$\text{Feasible set: } \mathcal{Z}(x_0) := \left\{ Z, V \left| \begin{array}{ll} z_{i+1} = Az_i + Bv_i & i \in [0, N-1] \\ z_i \in \mathcal{X} \ominus \mathcal{E} & i \in [0, N-1] \\ v_i \in \mathcal{U} \ominus K\mathcal{E} & i \in [0, N-1] \\ z_N \in \mathcal{X}_f \\ x_0 \in z_0 \oplus \mathcal{E} \end{array} \right. \right\}$$

$$\text{Cost function: } J(Z, V) := \sum_{i=0}^{N-1} l(z_i, v_i) + l_f(z_N)$$

$$\text{Optimization: } (V^*(x_0), Z^*(x_0)) = \underset{V, Z}{\operatorname{argmin}} \{ J(Z, V) \mid (Z, V) \in \mathcal{Z}(x_0) \}$$

$$\text{Control law: } \mu_{\text{tube}}(x) := K(x - z_0^*(x)) + v_0^*(x)$$

- Optimizing the nominal system, with tightened state and input constraints
- First tube center is optimization variable $\rightarrow$ has to be within $\mathcal{E}$ of x_0
- The cost is with respect to the tube centers (nominal system)
- The terminal set is with respect to the tightened constraints

Tube MPC

What do we need to make this work?

- Compute the set $\mathcal{E}$ that the error will remain inside
- Modify constraints on nominal trajectory $\{z_i\}$
- Formulate as convex optimization problem

... and then prove that

- **Constraints are robustly satisfied**
- The closed-loop system is robustly stable

Tube MPC Assumptions

Much the same as for nominal MPC:

1. The stage cost is a positive definite function, i.e. it is strictly positive and only zero at the origin
2. The terminal set is invariant **for the nominal system** under the local control law $\kappa_f(z)$:

$$Az + B\kappa_f(z) \in \mathcal{X}_f \quad \text{for all } z \in \mathcal{X}_f$$

All **tightened state and input constraints** are satisfied in $\mathcal{X}_f$:

$$\mathcal{X}_f \subseteq \mathcal{X} \ominus \mathcal{E}, \kappa_f(z) \in \mathcal{U} \ominus K\mathcal{E} \quad \text{for all } z \in \mathcal{X}_f$$

3. Terminal cost is a continuous Lyapunov function in the terminal set $\mathcal{X}_f$:

$$l_f(Az + B\kappa_f(z)) - l_f(z) \leq -l(z, \kappa_f(z)) \quad \text{for all } z \in \mathcal{X}_f$$

Robust Invariance

Thm: Robust Invariance of Tube MPC

The set $\mathcal{Z} := \{x \mid \mathcal{Z}(x) \neq \emptyset\}$ is a robust invariant set of the system $x(k+1) = Ax(k) + B\mu_{\text{tube}}(x(k)) + w(k)$ subject to the constraints $x, u \in \mathcal{X} \times \mathcal{U}$.

Let $(\{v_0^*, \dots, v_{N-1}^*\}, \{z_0^*, \dots, z_N^*\})$ be the optimal solution for $x(k)$.

At the next point in time, the state is:

$$x(k+1) = Ax(k) + BK(x(k) - z_0^*) + Bv_0^* + w \quad \text{for some } w \in \mathcal{W}$$

i.e., the state $x(k+1)$ may have many possible values. We need to show that there exists a feasible solution for **all of them**.

By construction, the state $x(k+1)$ is in the set $z_1^* \oplus \mathcal{E}$ for all $\mathcal{W}$. Therefore (as in standard MPC), the sequence

$$(\{v_1^*, \dots, v_{N-1}^*, \kappa_f(z_N^*)\}, \{z_1^*, \dots, z_N^*, Az_N^* + B\kappa_f(z_N^*)\})$$

is feasible for all $x(k+1)$.

Tube MPC

What do we need to make this work?

- Compute the set $\mathcal{E}$ that the error will remain inside
- Modify constraints on nominal trajectory $\{z_i\}$
- Formulate as convex optimization problem

... and then prove that

- Constraints are robustly satisfied
- **The closed-loop system is robustly stable**

Robust Stability

Thm: Robust Stability of Tube MPC

The state $x(k)$ of the system $x(k+1) = Ax(k) + B\mu_{\text{tube}}(x(k)) + w(k)$ converges in the limit to the set $\mathcal{E}$.

As in standard MPC, we have the relationship:

$$\begin{aligned} J^*(x(k)) &= \sum_{i=0}^{N-1} l(z_i^*, v_i^*) + l_f(z_N^*) \\ J^*(x(k+1)) &\leq \sum_{i=1}^N l(z_i^*, v_i^*) + l_f(z_{N+1}) \\ &= J^*(x(k)) - \underbrace{l(z_0^*, v_0^*)}_{\geq 0} - \underbrace{l_f(z_N^*) + l_f(z_{N+1}) + l(z_N^*, \kappa_f(z_N^*))}_{\leq 0 \text{ (} l_f \text{ is a Lyapunov function in } \mathcal{X}_f \text{)}} \end{aligned}$$

This shows that $\lim_{k \rightarrow \infty} J(z_0^*(x(k))) = 0$, and therefore $\lim_{k \rightarrow \infty} z_0^*(x(k)) = 0$.

However, $x(k)$ does not tend to zero! It only stays within a robust invariant set centered at $z_0^*(x(k))$: $\lim_{k \rightarrow \infty} \text{dist}(x(k), \mathcal{E}) = 0$, where dist is any distance function.

Putting it all together: Tube MPC

To implement tube MPC:

— Offline —

1. Choose a stabilizing controller K so that $A + BK$ is (Schur) stable
2. Compute the minimal robust invariant set $\mathcal{E} = \mathcal{F}_\infty$ for the system $x(k+1) = (A + BK)x(k) + w(k)$, $w \in \mathcal{W}^1$
3. Compute the tightened constraints $\tilde{\mathcal{X}} := \mathcal{X} \ominus \mathcal{E}$, $\tilde{\mathcal{U}} := \mathcal{U} \ominus K\mathcal{E}$
4. Choose terminal weight function l_f and constraint $\mathcal{X}_f$ satisfying assumptions on slide 40

— Online —

1. Measure / estimate state x
2. Solve the problem $(V^*(x), Z^*(x)) = \operatorname{argmin}_{V,Z} \{J(Z, V) \mid (Z, V) \in \mathcal{Z}(x)\}$ (Slide 38)
3. Set the input to $u = K(x - z_0^*(x)) + v_0^*(x)$

¹Note that it is often not possible to compute the minimal robust invariant set, as it may have an infinite number of facets. Therefore, we often take an invariant outer approximation.

Example

System dynamics

$$x(k+1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x(k) + \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} u(k) + w(k)$$

$$\mathcal{W} := \{w \mid |w_1| \leq 0.01, |w_2| \leq 0.1\}$$

Constraints:

$$\mathcal{X} := \{x \mid \|x\|_\infty \leq 3\}$$

$$\mathcal{U} := \{u \mid \|u\| \leq 0.5\}$$

Stage cost is:

$$l(z, v) := z^\top Q z + v^\top R v$$

where

$$Q := \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

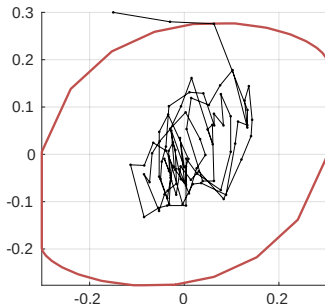
$$R := 10$$

Offline Design - Compute Minimal Invariant Set

1. Choose a stabilizing controller K so that $A + BK$ (Schur) stable
2. Compute the minimal robust invariant set $\mathcal{E} = \mathcal{F}_\infty$ for the system $x(k+1) = (A + BK)x(k) + w(k)$, $w \in \mathcal{W}$

We take the LQR controller for $Q = I$, $R = 10$:

$$K := \begin{bmatrix} -0.2311 & -0.7011 \end{bmatrix}$$



Evolution of the system
 $x(k+1) = (A+BK)x(k) + w(k)$
for $x(0) = \begin{bmatrix} -0.15 & 0.3 \end{bmatrix}^T$

Offline Design - Tighten State Constraints

3. Compute the tightened constraints $\tilde{\mathcal{X}} := \mathcal{X} \ominus \mathcal{E}$, $\tilde{\mathcal{U}} := \mathcal{U} \ominus K\mathcal{E}$

$$\mathcal{X} = \{x \mid \|x\|_{\infty} \leq 3\} = \left\{x \mid \begin{bmatrix} I \\ -I \end{bmatrix} x \leq 3 \begin{bmatrix} \mathbf{1} \\ \mathbf{1} \end{bmatrix}\right\}$$

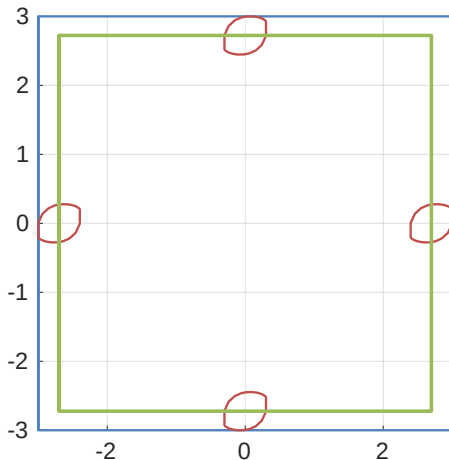
If $\mathcal{E} = \{x \mid Fx \leq f\}$, then the tightened constraint sets are:

$$\begin{aligned} \mathcal{X} \ominus \mathcal{E} &= \{x \mid x + e \in \mathcal{X} \ \forall e \in \mathcal{E}\} = \left\{x \mid \begin{bmatrix} I \\ -I \end{bmatrix} x + \begin{bmatrix} I \\ -I \end{bmatrix} e \leq 3 \begin{bmatrix} \mathbf{1} \\ \mathbf{1} \end{bmatrix} \ \forall e \in \mathcal{E}\right\} \\ &= \left\{x \mid \begin{bmatrix} I \\ -I \end{bmatrix} x \leq \begin{bmatrix} 3 - \max \left\{ \begin{bmatrix} 1 & 0 \end{bmatrix} e \mid e \in \mathcal{E} \right\} \\ 3 - \max \left\{ \begin{bmatrix} 0 & 1 \end{bmatrix} e \mid e \in \mathcal{E} \right\} \\ 3 - \max \left\{ \begin{bmatrix} -1 & 0 \end{bmatrix} e \mid e \in \mathcal{E} \right\} \\ 3 - \max \left\{ \begin{bmatrix} 0 & -1 \end{bmatrix} e \mid e \in \mathcal{E} \right\} \end{bmatrix}\right\} \end{aligned}$$

The maximizations are all linear programs and can be computed offline.

The result is a polytope with smaller right hand side.

Offline Design - Tighten State Constraints



Blue : Original constraint set $\mathcal{X}$

Red : Error set $\mathcal{E}$

Green : Tightened constraints $\mathcal{X} \ominus \mathcal{E}$

Offline Design - Tighten Input Constraints

We compute $\mathcal{U} \ominus K\mathcal{E}$ in the same manner:

$$\begin{aligned}\mathcal{U} \ominus K\mathcal{E} &= \left\{ u \mid \begin{bmatrix} 1 \\ -1 \end{bmatrix} u \leq \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} \right\} \ominus \{Ke \mid Fe \leq f\} \\ &= \left\{ u \mid \begin{bmatrix} 1 \\ -1 \end{bmatrix} u \leq \begin{bmatrix} 0.5 - \max\{Ke \mid Fe \leq f\} \\ 0.5 + \min\{Ke \mid Fe \leq f\} \end{bmatrix} \right\}\end{aligned}$$

Offline Design - Terminal Weights and Constraints

We need to find a function l_f and a set $\mathcal{X}_f$ that satisfy the conditions on slide 40:

1. The terminal set is invariant **for the nominal system** under the local control law $\kappa_f(z)$:

$$Az + B\kappa_f(z) \in \mathcal{X}_f \quad \text{for all } z \in \mathcal{X}_f$$

All **tightened state and input constraints** are satisfied in $\mathcal{X}_f$:

$$\mathcal{X}_f \subseteq \mathcal{X} \ominus \mathcal{E}, \kappa_f(z) \in \mathcal{U} \ominus K\mathcal{E} \quad \text{for all } z \in \mathcal{X}_f$$

2. Terminal cost is a continuous Lyapunov function in the terminal set $\mathcal{X}_f$:

$$l_f(Az + B\kappa_f(z)) - l_f(z) \leq -l(z, \kappa_f(z)) \quad \text{for all } z \in \mathcal{X}_f$$

Offline Design - Terminal Constraint

We base our terminal weights and constraints on the LQR controller (many other choices possible).

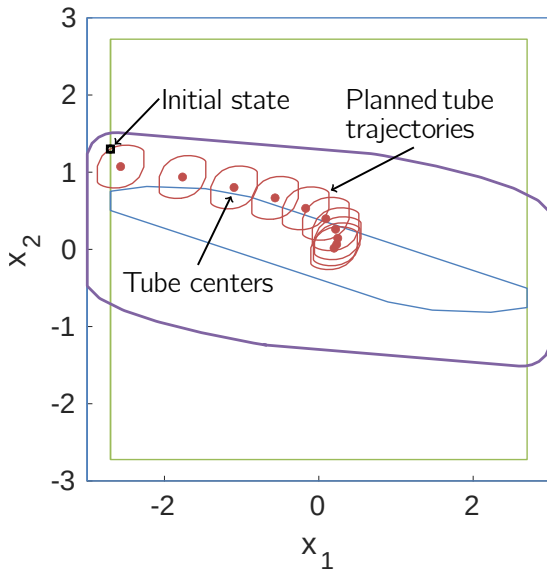
- Choose the terminal control law to the LQR control law: $\kappa_f(x) = Kx$ where the weights Q and R are taken the same as for our MPC problem.
- We need a set $\mathcal{X}_f$ that is invariant under this controller and contained in the tightened constraints. Can be obtained similarly as in the nominal MPC case.
- Recall the optimal cost of the LQR control law is:

$$J^*(z_0) = \sum_{i=0}^{\infty} z_i^\top (Q + K^\top R K) z_i = z_0^\top P z_0$$

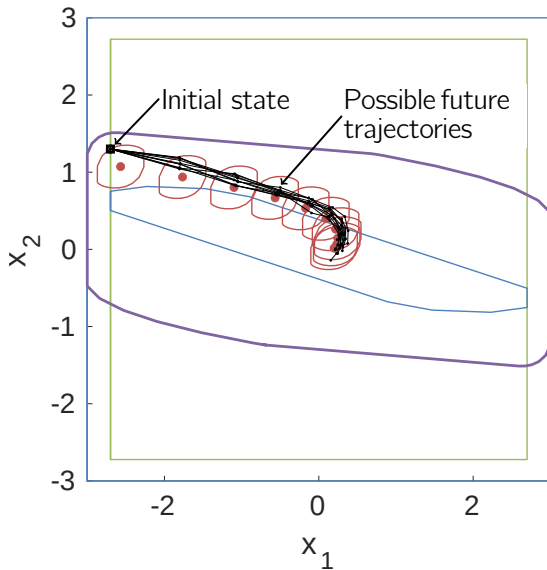
where P is the solution to a discrete-time Riccati equation.

We know that $J^*(z)$ is a Lyapunov function for $z(k+1) = (A + BK)z(k)$ and we can take $l_f(z) = z^\top P z$.

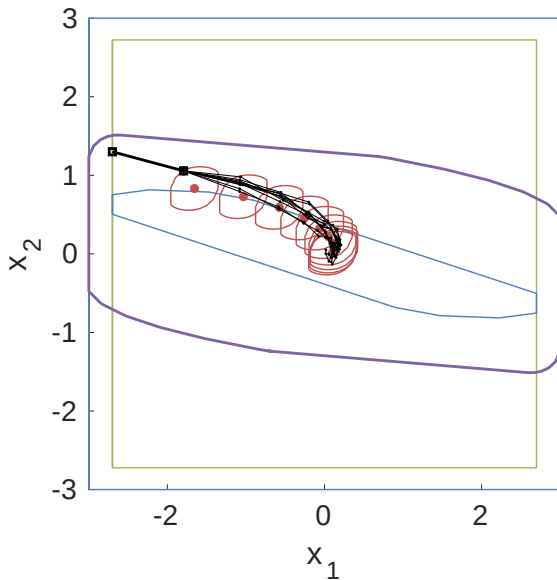
Tubes - Example



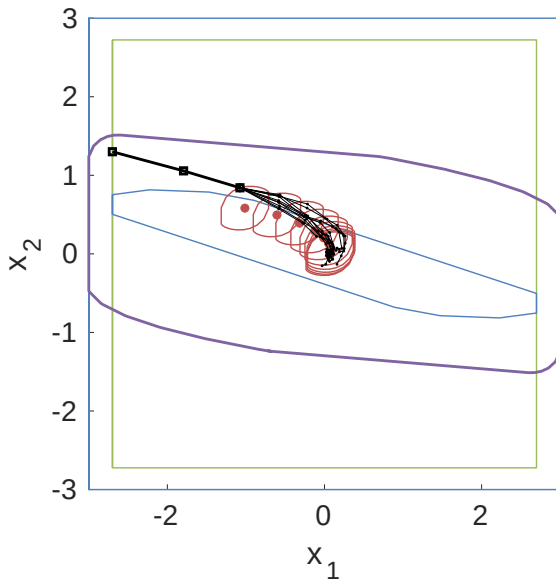
Tubes - Example



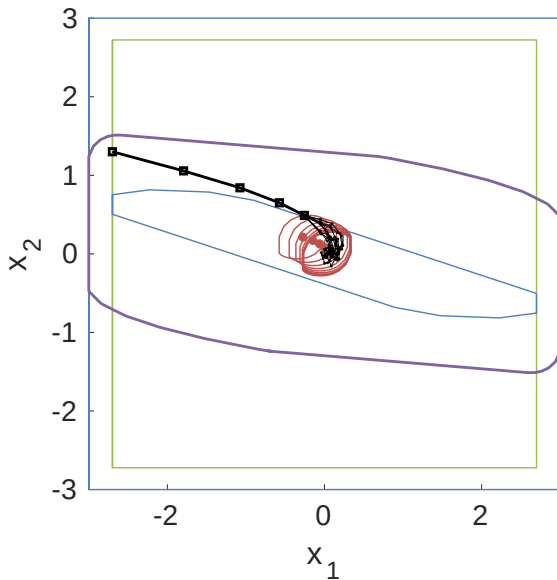
Tubes - Example



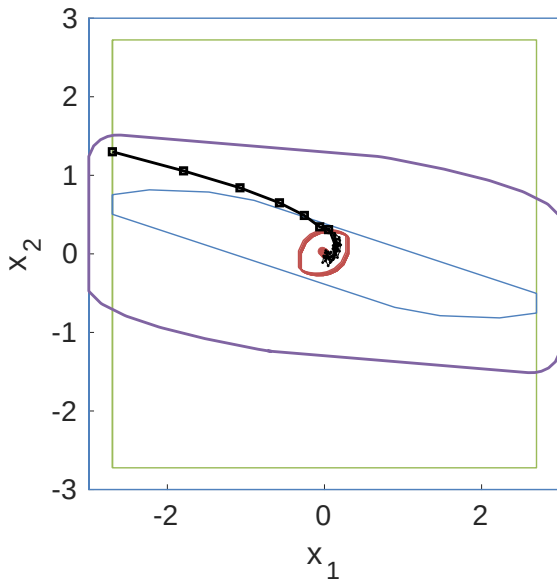
Tubes - Example



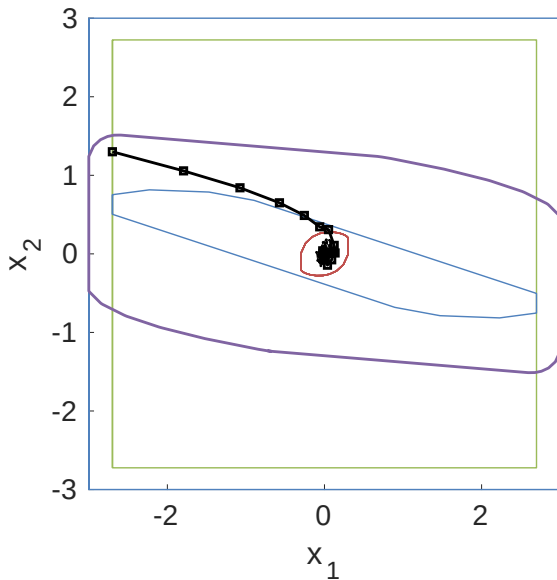
Tubes - Example



Tubes - Example



Tubes - Example



Tube MPC - Summary

Idea:

- Split input into two parts: One to steer nominal system (v), one to compensate for the noise (Ke): $u = Ke + v$
- Optimize for the nominal trajectory, ensuring that any deviations stay within constraints

Benefits:

- Compute one constraint tightening
- Optimization problem to solve is simple
- Same terminal set and cost design as in nominal MPC
- Nominal MPC theory applies to prove convergence of nominal state

Cons:

- Sub-optimal MPC (optimal is extremely difficult)
- Reduced feasible set when compared to nominal MPC
- We need to know what $\mathcal{W}$ is (this is usually not realistic)

Robust MPC for uncertain systems

Robust MPC for uncertain systems

We described two robust MPC approaches:

Constraint-tightening MPC

- Disturbance reachable sets (DRS) $\mathcal{F}_k$

⇒ Nominal MPC with tightened constraints $\mathcal{X} \ominus \mathcal{F}_i$ initial state $z_0 = x(k)$.

Tube-MPC

- Robust positive invariant (RPI) set $\mathcal{E}$

⇒ Nominal MPC with tightened constraints $\mathcal{X} \ominus \mathcal{E}$ initial state $z_0 \in x(k) \oplus \mathcal{E}$.

Robust MPC for uncertain systems

We described two robust MPC approaches:

Constraint-tightening MPC

- Disturbance reachable sets (DRS) $\mathcal{F}_k$
- ⇒ Nominal MPC with tightened constraints $\mathcal{X} \ominus \mathcal{F}_i$ initial state $z_0 = x(k)$.

Tube-MPC

- Robust positive invariant (RPI) set $\mathcal{E}$
- ⇒ Nominal MPC with tightened constraints $\mathcal{X} \ominus \mathcal{E}$ initial state $z_0 \in x(k) \oplus \mathcal{E}$.

Discussion

- RPI set $\mathcal{E}$ increases complexity of the offline design and online implementation.
- Tube-MPC has a smaller feasible set.
- Tube-MPC has simple concept with strong stability properties

Robust MPC for Uncertain Systems - Summary

Idea

- Compensate for noise in prediction to ensure all constraints will be met

Cons

- Complex (some schemes are simple to implement, like tubes, but complex to understand)
- Must know the largest noise $\mathcal{W}$
- Often conservative
- Feasible set may be small

Benefits

- Feasible set is invariant - we know exactly when the controller will work
- Easier to tune - knobs to tradeoff robustness against performance